Unknown surveillance: one fixed observed dataset under 16 detection assumptions

Data-generating truth: R_{eff}=0.50, k=0.15, p_p=0.25, p_a=0.50; n=5000 observed clusters

Joint 2-D 95% contour | Inferred pair (MLE) | True transmission pair | Independent 1-D 95% profile CIs

(a) p_p=0.25, p_a=1.00 | (b) p_p=0.50, p_a=1.00 | (c) p_p=0.75, p_a=1.00 | (d) p_p=1.00, p_a=1.00

(e) p_p=0.25, p_a=0.75 | (f) p_p=0.50, p_a=0.75 | (g) p_p=0.75, p_a=0.75 | (h) p_p=1.00, p_a=0.75

(i) p_p=0.25, p_a=0.50 | (j) p_p=0.50, p_a=0.50 | (k) p_p=0.75, p_a=0.50 | (l) p_p=1.00, p_a=0.50

(m) p_p=0.25, p_a=0.25 | (n) p_p=0.50, p_a=0.25 | (o) p_p=0.75, p_a=0.25 | (p) p_p=1.00, p_a=0.25

Effective reproduction number, R_{eff}

0.7
0.6
0.5
0.4
0.3

0.2
0.4
0.6
0.8

Dispersion parameter, k (linear scale)